AA-C-001771C(GSA-FSS)
July 23, 1979
SUPERSEDING
Int. Fed. Spec. AA-C-001771B(GSA-FSS)
August 1, 1975

INTERIM FEDERAL SPECIFICATION

CHAIRS, DOUBLE SHELL, MOLDED PLASTIC, ROTARY, UPHOLSTERED

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers upholstered chairs of the molded plastic double shell design of contemporary style for use in offices, conference rooms, and other similar applications.

1.2 Classification.

1.2.1 Types and styles. Chairs covered by this specification shall be of the following types and styles, as specified (see 6.2).

Type I - High back desk chair, rotary, tilting seat and back, adjustable seat height, casters, tufted upholstery.

Style A - with arms.

Type II - Low back desk chair, rotary, tilting seat and back, adjustable seat height, casters, tufted upholstery.

Style A - with arms.
Style B - without arms.

Type III - Side chair, rotary, nontilting seat and back, fixed seat height, glides, tufted upholstery.

Style A - with arms.
Style B - without arms.

Type IV - Secretarial chair, rotary, nontilting seat, tilt back, adjustable seat height, adjustable back, casters, tufted upholstery.

Style B - without arms.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

FF-C-82 - Casters, Swivel, Chair.
QQ-A-591 - Aluminum Alloy Die Castings.
QQ-R-571 - Rods, Welding, Copper and Nickel Alloys.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Association of Textile Chemists and Colorists:

Test Method 16A - Colorfastness to Light: Carbon-Arc Lamp, Continuous Light.
Test Method 16E - Colorfastness to Light: Water-Cooled Xenon-Arc Lamp, Continuous Light.

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

American Society for Testing and Materials (ASTM) Standards:

B 117 - Salt Spray (Fog) Testing.
D 1175 - Abrasion Resistance of Textile Fabrics.
D 1564 - Tentative Methods of Test for Flexible Urethane Foam.
D 2406 - Molded Flexible Urethane Foam.
D 3597 - Woven Upholstery Fabrics - Plain, Tufted, or Flocked.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Iron and Steel Institute, Publication:

Steel Products Manual, Carbon Steel Sheets.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, NY 10017.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Application of preproduction samples. There shall be three chairs and a complete set of component parts (base with casters or glides, chair control, shells, polyurethane foam and fabric) of each type and style on contract used in the preproduction examination. One of the preproduction sample chairs shall be used for the dimensional examination in table II and the packaging and packing examination in Section 5. The other two preproduction sample chairs and component parts shall be used for all other examinations and tests.

3.2 Material. All materials used shall be as hereinafter specified. They shall be free from defects which affect serviceability or appearance of the finished product. The end item(s) shall contain not less than 7 percent, by weight, of recovered materials, excluding packaging. Recovered materials are defined as materials which have been collected or recovered from solid waste.

3.2.1 Steel. The steel sheets shall be of Commercial Quality cold rolled steel as defined in the Steel Products Manual, Carbon Steel Sheets. The steel shall be smooth, free from rust, scale, pits, scratches, laps, and buckles affecting strength or appearance. Tubular steel shall be Commercial Quality SAE 1010-1020, seamless or continuously welded cold rolled steel. Alternatively, tubular steel may be number 434 high tensile, satin finished, stainless steel. Unless otherwise specified herein, the outside corners of all tubing shall be rounded.

3.2.2 Polyurethane foam. Polyurethane foam shall conform to the requirements of table I when tested as specified in the applicable sections of ASTM Methods D 1564 or D 2406. The contractor shall submit adequately documented test data from an independent testing laboratory showing that the polyurethane foam complies with the requirements in table I.

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TABLE I. Physical properties of polyurethane - slab and molded foam (tested according to ASTM D 1564 or ASTM D 2406 as applicable)

Property	Low Density	Medium Density	High Density
25 percent indentation and deflection (lbs) 1/	14 - 31	31 - 37	37 - 64
Indentation load ratio (65 percent/25 percent) minimum	1.8	1.9	1.9
Original compression set 22 hr/158° F, deflected to 10 percent of its thickness, percent maximum	20 percent	15 percent	15 percent
Fatigue test 25 percent indentation load ratio loss, percent maximum	30 percent	30 percent	25 percent
Maximum thickness loss	5 percent	5 percent	5 percent
Density (lbs/cu ft)	1.4 minimum	2.7 - 3.4	2.7 - 4.8

1/ In those cases where foams having thicknesses of 4 inches are not available, the following reduced ILD values shall be used:

3 inches - 90 percent of the 4-inch ILD value.
2 inches - 80 percent of the 4-inch ILD value.
1 inch - 70 percent of the 4-inch ILD value.

3.2.3 Upholstery fabric. The upholstery fabric shall be 100 percent flat nylon, dyed, plain weave, not less than 21 ounces per linear yard (based on 54 inch width) exclusive of back coating. The fabric shall conform to the requirements of table I of ASTM D 3597 except: the minimum breaking strength and tear strength shall be 250 lbs and 20 lbs, respectively, for warp and filling; and the surface abrasion test shall be the Taber Abrader method (with CS#10 wheels, 500 gram load) in accordance with ASTM D 1175 and the fabric shall withstand 8,000 revolutions without a complete breakthrough of a yarn. The colorfastness to light test shall be conducted in accordance with AATCC 16A (AATCC 16E is an acceptable alternate with an exposure of 50 hours).

(In ASTM D 3597 disregard Note 6 concerning relevancy of test if fabric is to be labeled "clean with solvent only" and paragraphs 7.6.2 through 7.6.4). The fabric shall have a seam efficiency not less than 80 percent when tested under method 5110 of Federal Test Method Standard No. 191, with needle size and thread type and size the same as or comparable to that which would be used in commercial production, or stronger (nylon thread is acceptable for this test).

3.2.3.1 Color of upholstery fabric. The color of the upholstery fabric shall be as specified by the contracting officer (see 6.2). The colors specified shall match the Government control samples listed below (see 6.3).

Orange/Rust	FSS-F-01011
Sand/White	FSS-F-01015
Blue/Black	FSS-F-01005
Black/White	FSS-F-01014
Black/Brown	FSS-F-01012
Red/Black	FSS-F-01013
Green/Black	FSS-F-01016
Yellow/Gold	FSS-F-01018
Black	FSS-F-01009

3.2.4 Finishing materials.

3.2.4.1 Painted finish.

3.2.4.1.1 Primer. The primer, when used, shall be compatible with the finishing enamel.

3.2.4.1.2 Enamel. The enamel for the finishing shall be of the baking type in accordance with the manufacturer's commercial product.

3.2.4.2 Chromium plating. The materials used in the plating process shall be as normally used commercially to accomplish the quality specified herein (see 3.5).

3.2.5 Hardware.

3.2.5.1 Fastening devices. Unless otherwise specified herein, all bolts, nuts, screws, washers, and other fastening devices shall be of the type, style, and size normally used commercially for the purpose intended and shall be made corrosion resistant in accordance with the manufacturer's commercial practice, except when paint or lacquer is involved. The use of paint or lacquer as the sole corrosion-resistant agent is prohibited.

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3.2.5.2 Casters. The casters shall be type I, style A, of FF-C-82 with a 3/8 inch or 7/16 inch diameter stem. The wheel shall be 2-1/2 inches in diameter with rubber tread material having a Shore Durometer "A" scale hardness of 85 to 95 or equivalent on a comparable scale. The finish of the exterior metal surfaces of the caster shall match the plated finish used on the base within the limits of the caster manufacturer's commercial finish for executive chairs. Alternatively, when the base is provided as specified in 3.4.3(b), the finish of the exterior metal surface may be provided in black epoxy finish to match the black finish of the vertical sides of the base.

3.2.5.3 Glides. The glides shall be polished nickel plated, polished chromium plated or polished stainless steel. They shall be of the live-rubber-cushioned spring steel expansion type, rubber expansion type, PVC cushioned type or metal expansion ring pintal type. The glides shall be firmly secured to each leg and shall be capable of removal without the use of special tools. There shall be no metal-to-metal contact of glide base and base leg after the glides are applied. The glides shall have a diameter of not less than 15/16 inch. The glides shall have a substantially flat bottom. The base of the glide shall be stamped from not less than 0.030 inch thick steel and shall be case hardened after forming to a Rockwell hardness of not less than 15 N 85 or equivalent on a comparable scale. The base of the glide shall have round edges.

3.2.5.4 Chair controls.

3.2.5.4.1 Chair controls, type I, type II chairs. The chair control mechanism shall have a low profile, extending not more than 3-1/2 inches below the bottom surface of the outer shell. The chair controls shall provide for seat height adjustment over a range of not less than 2-1/4 inches, revolving of the seat over a full 360 degrees, seat tilt over a range of not less than 16 degrees or more than 24 degrees (see 4.4.2.1) and seat tilt tension adjustment. Seat height adjustment shall be accomplished by a spindle cover, a tool clipped to a holding device permanently attached to the underside of the seat or by rotating the unoccupied seat. The seat shall not change height unintentionally when occupied. Tilt tension adjustment shall be accomplished by a single visible hand hold or a tool as described above. All applicable parts of the mechanism shall be properly lubricated. All parts shall have a painted finish as specified in 3.5 and the color shall be black, except that the spindle cover shall be plated as specified in 3.5.

3.2.5.4.2 Seat tilting unit, type I and type II chairs. The tilting unit shall provide the tension against the tilting of the seat by means of springs, rubber torsion units, steel torsion rods, or any other means having commercial acceptance that allows for variable tension. Tilting tension bolts shall be not less than 3/8 inch nominal diameter and must be entirely supported by the chair control structure, not the seat. The bolt shall have ample clearance through the entire tilting range to prevent breaking or wear by friction. The tilting unit shall have a positive safety stop that will limit the tilting action during normal use and in case of failure of the tension unit. The seat tilting mechanism shall be permanently lubricated or equipped with nylon bearings and/or oil impregnated, powdered metal bearings to prevent wear on bearing surfaces.

3.2.5.4.3 Chair controls, type III chairs. The type III chair shall rotate a full 360 degrees or shall allow the chair seat to rotate not less than 320 degrees and the chair seat shall automatically return to its original position when the occupant leaves the seat. The chair control shall have a low profile, extending not more than 3-1/2 inches below the bottom surface of the outer shell. The spindle shall be completely covered by the base hub or by a cylindrical cover. The spindle cover shall be chromium plated as specified in 3.5. This control shall not have seat tilting action or seat height adjustment. All applicable parts of the mechanism shall be properly lubricated.

3.2.5.4.4 Chair controls, type IV chairs. The chair control mechanism shall have a low profile, extending not more than 3-1/2 inches below the bottom surface of the outer shell. The chair controls shall provide for seat height adjustment over a range of not less than 2-1/4 inches, revolving of the seat over a full 360 degrees, a nontilt seat, a back upright tilt of not more than 17 degrees (see 4.4.2.1), back upright tilt tension adjustment, a horizontal (front to rear) adjustment of the back upright not less than 1-1/4 inches and vertical adjustment of the back not less than 2 inches along the back upright. Seat height adjustment shall be accomplished by a spindle cover, a tool clipped to a holding device permanently attached to the underside of the seat or by rotating the unoccupied seat. The seat shall not change height unintentionally when occupied. Back upright tilt tension adjustment shall be accomplished by a single visible hand hold or a tool as described above. Provision shall be made to prevent the accidental removal of the back upright and back. All applicable parts of the mechanism shall be properly lubricated. All parts shall have a painted finish as specified in 3.5 and the color shall be black, except that the spindle cover and back upright shall be plated as specified in 3.5.

3.2.5.4.5 Back tilting unit, type IV chairs. The back tilting unit shall provide the tension against the tilting of the chair back by means of springs, rubber torsion units, steel torsion rods or any other means having commercial acceptance that allows for variable tension. Tilting tension bolts shall be not less than 3/8 inch nominal diameter and must be entirely supported by the chair control structure. The bolts shall have ample clearance through the entire tilting range to prevent breakage or wear by friction. The tilting action shall have a positive safety stop that will limit the tilting action during normal use and in case of failure or breakage of the tension unit. Provision shall be made in the tilting mechanism to reduce the shock to the occupant when the chair is tilted to the full extent of its range. All applicable parts of the mechanism shall be properly lubricated.

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3.2.5.4.6 Chassis, spider, and spindle. The parts shall be securely assembled and shall provide rigid and properly balanced support between the chair seat and the base within specified adjustment limitations. The spindle shall be cold drawn steel 1 inch (plus 0.000 inch, minus 0.002 inch) in diameter and permanently attached to the chassis with the keyway slot facing to the side (not forward or rearward). Threads on spindles shall terminate at a minimum of 2-1/2 inches from the bottom end, except when twin hub bearings are used in the base, the threads shall then terminate at a minimum of 3/8 inch from the bottom end. Alternatively, on a rolled thread spindle, the spindle may be fully threaded but the keyway slot shall terminate at a minimum of 2-1/2 inches from the bottom end of the spindle and a steel "J" washer shall be used. The threads and the keyway slots on all spindles shall be free of burrs and sharp edges. The chassis shall extend a minimum of 3/4 inch along the spindle and be designed to receive and give support to the spindle. The spider shall be steel, and shall be rigidly secured to the chair shell in not less than four places by bolts, nuts and lockwashers, bolts and locknuts, or by bolts and nuts locked together by use of a chemical locking compound. The spindle shall be enclosed its full length with not more than 1/2 inch protrusion below the base when in the low position by a circular metal cover or base hub. When in the high position, the spindle shall be enclosed by the cover or base hub so that not more than 1 inch of the spindle is visible. The seat shall not change height unintentionally during normal in-use rotation. The spindle shall be prevented from being accidently removed from the base.

3.2.5.4.7 Replaceable "J" washers. If a "J" washer is used, the seat height adjustment control mechanism shall allow for the later insertion of plastic "J" washers which are designed either to open in order to fit over the spindle or which incorporate a removable metal part which performs as a key. The insertable "J" washers shall be obtainable by the chair users through the efforts of the chair manufacturer. Each chair shall have a label permanently affixed to the underside of the chair seat which states that, when needed, a replacement insertable "J" washer can be obtained by contacting the chair manufacturer for a source.

3.3 Design, dimensions.

3.3.1 Design. The basic design and general appearance of the chairs covered by this specification shall be illustrated in the figures identified below. The figures are not intended to preclude similar chairs which are otherwise in accordance with the requirements of this specification.

Chair	Figure
Type I, style A	1
Type II, style A	2
Type II, style B	2
Type III, style A	3
Type III, style B	3
Type IV, style B	4

3.3.2 Dimensions. The dimensions of the chairs shall be as shown in table II.

TABLE II. Dimensions (inches) 1/

	Type I	Type II		Type III		Type IV
	Style A	Style A	Style B	Style A	Style B	Style B
A. Seat depth	17 - 20	17 - 20	17 - 20	17 - 20	17 - 20	16 - 19
B. Seat height 2/ (No adjustment) 3/	-------	-------	-------	16 - 18	16 - 18	-------
Seat height 2/ (Lowest adjustment setting)	17 - 18	17 - 18	17 - 18	-------	-------	16 - 18
C. Overall height (No adjustment) 3/	-------	-------	-------	30 - 33	30 - 33	-------
Overall height 4/ (Lowest adjustment setting)	36 - 44	30 - 34	30 - 34	-------	-------	-------
D. Center to center caster or glide pintle	25 - 29	25 - 27	25 - 27	25 - 27	25 - 27	22 - 25
E. Width between arms	20 min.	20 min.	-------	20 min.	-------	-------
F. Width of seat	-------	-------	19 - 24	-------	19 - 24	16 - 20
G. Width of chair back	-------	-------	-------	-------	-------	14 - 18
H. Height of chair back	-------	-------	-------	-------	-------	9 - 11

1/ Points at which measurements are taken are illustrated in figure 5.
2/ Measured from a horizontal line along the front center to rear center of the chair seat intersecting the highest point of the seat on that line.
3/ With glides in base (not casters).
4/ There shall be a minimum 4 inch difference in overall height between the type I and type II chairs of the same manufacturer.

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3.4 Construction. Any detail not specified herein shall be consistent with good quality production of similar commercial products. All elements of construction shall be free of any conditions which hinder the proper function of any part of the chairs or which constitute a potential hazard to personnel or their clothing. Components and parts shall be properly aligned to assure neat appearance of the finished chair.

3.4.1 Assembly of metal parts. The assembly of metal parts shall be by welding, brazing, or mechanical attachments as specified herein for maximum strength at each joint. Welding and brazing shall be sound and without porosity and shall provide rigid chairs that have sufficient strength to prohibit the loosening of any part; or shall not cause any other defect which might affect the service life of the chairs. Exposed welds shall be ground smooth to present a neat finished appearance. All overlapping parts which are to be welded to each other shall be positioned to provide maximum welding area, strength and rigidity to the welded assembly. Mechanical attachments shall provide secure and rigid joints in proper alignment. Where not otherwise specified, mechanical joints shall be made with cadmium or zinc plated steel rivets, bolts, nuts, and lockwashers.

3.4.2 Brazing. Where brazing is permitted, the following conditions shall apply: When brazing is used to join component parts in lieu of welding, the brazing method shall include the use of a nickel-bronze rod conforming to type I, class RCuZn - D of QQ-R-571, and shall have a tensile strength of 70,000 p.s.i., or, alternatively, silver brazing with 45 percent silver content may be used in conjunction with an induction welding process. When brazing is used simply as a filler or a filler to hide a seam or joint that has been securely joined by welding, the brazing method may include the use of any rod having adequate tensile strength to assure the brazed area will not crack or fracture under shipping, moving, or in-use conditions.

3.4.3 Base. The base shall be modern contemporary design, constructed as specified in (a), (b), or (c). The base hub shall provide adequate structural support to the spindle to prevent wobble and instability. The hub and spindle shall be properly lubricated to prevent wear and noise during service.

(a) The design of the base shall be generally rectangular in profile. The legs shall be made of steel tubing or formed steel welded into a tube. The end of the leg shall be closed with a metal insert which is securely welded or induction welded and brazed to the leg with the joint ground smooth to eliminate a visible joint. The legs shall be securely welded to the hub and/or hub and spindle cover. All joints shall be finished to eliminate open joints exposed to normal view. The base leg shall be sloped downward (from the point of attachment to the base hub to the end of the base leg) from a minimum of 1/4 inch to a maximum of 2-1/4 inches on the top surface, and from a minimum of no slope to a maximum of 1-1/4 inches on the bottom surface. All outer surfaces of the base, including spindle cover, shall be chromium plated in accordance with 3.5.

(b) The base shall be constructed of steel and shall be a slim-line "T" shape design. The "T" shape legs shall be continously welded (on both sides of each leg) to a center, circular hub. The leg-hub assembly shall have a painted finish as specified in 3.5 in black color. The ends of the base legs shall have bumpers or toe caps of molded ABS plastic or nylon in black color. The top surface of the base or top of the "T" shape design shall be chromium plated in accordance with 3.5 and extending from the top-front of each leg to the top of the center column (hub).

(c) The base shall be die cast aluminum (conforming to alloy 380 of QQ-A-591) of modern contemporary design similar to the designs indicated in subparagraphs (a) or (b) above. The base shall have a decorative satin finish which shall visually match Government sample number FSS-M-01001 (see 6.3).

3.4.3.1 Caster and glide sockets. Caster sockets to accommodate the casters specified in 3.2.5.2 shall be provided in bases for all type I, type II, and type IV chairs. Sockets to accommodate the glides specified in 3.2.5.3 shall be provided in bases for all type III chairs. The caster and glide sockets shall be accurately located to provide a true vertical position for the caster or glide and shall provide sufficient metal support at the end of the leg to insure secure caster or glide attachment at all times. The socket shall be securely welded or induction welded and brazed to the base. Alternatively, the socket may be an integral part of the steel base and insert or formed into a die cast aluminum base.

3.4.4 Shells.

3.4.4.1 Shells, type I, II, and III chairs. The chair seat and back shall consist of inner and outer molded plastic shells. The inner and outer shells shall be molded from plastics used in production units which have been sold previously in substantial quantities by the offerer and which have a proven performance in this application. The inner shell shall have provisions for tufting the padding and upholstery cover. The outer shell shall have a textured finish on the outside surface and shall be colored black or slate gray. The color of the outer shell shall be incorporated into the shell during molding or shall be colored with a baked-on polyester finish having a minimum dry film thickness of 2 mils. There shall be no holes or openings molded into the outer shell in the chair back area.

3.4.4.2 Shells, type IV chairs. The chair seat and back shall consist of a separate molded plastic outer shell and an inner shell. The molded plastics, when used, shall be as required in 3.4.4.1 above.

The inner shell shall be plastic; minimum five ply plywood (with edges rounded) having a minimum 3/8 inch seat and 1/4 inch back; or laminated jute fiberboard in the back. The inner shell shall have provision for tufting the padding and upholstery cover. The outside surface of the outer shell shall have a textured finish colored black or slate gray and applied as specified in 3.4.4.1.

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3.4.4.3 Spider and shell attachment. The chair control spider shall be securely attached to either the inner or outer shell by a method that will prevent accidental separation of spider from the shell and shall not cause damage to the shell. The inner and outer shells shall be securely fastened together to prevent separation, shifting, buckling, or gaps. When necessary to cover the joining of the two shells, the perimeter of the outer shell shall be covered with a black bumper strip of impact resistant flexible vinyl secured firmly to prevent accidental removal. All fastening devices shall remain tight and secure even after subjection to tests. The fastening devices shall not be visible from normal exterior view.

3.4.5 Arms. The arms shall be an integral part of the outer shell or shall be separate molded plastic components (with or without metal reinforcement) securely fastened to the inner shell so that the attaching brackets are not visible when the chair is upholstered. The arms shall be black with the color incorporated at the time of molding.

3.4.6 Seat and back padding. Either slab polyurethane or its equivalent in molded polyurethane may be used (see 3.2.2). Commercial tolerance of minus 1/16 inch for padding thickness shall apply.

3.4.6.1 Seat padding. The seat pad shall be high density polyurethane foam or a combination of high density polyurethane foam as the lower portion cemented to a low density polyurethane foam upper portion. The minimum thickness of the polyurethane foam shall be 2-1/2 inches for types I, II, and III chair and 1 inch for the type IV chair. Additional padding of any suitable material may be included at the option of the manufacturer. The seat pad shall be firmly secured to the inner shell by tufting and/or adhesive bonding.

3.4.6.2 Back padding. The upper portion of the back pad on types I, II, and III chairs shall be low density or medium density polyurethane foam with a minimum thickness of 1-3/4 inches. The lower portion of the back pad (lumbar support region) on types I, II, and III chairs and the back pad on the type IV chairs shall be polyurethane foam with a minimum thickness of 1-1/2 inches. Additional padding of any suitable material may be included at the option of the manufacturer. The back pad shall be firmly secured to the inner shell by tufting and/or adhesive bonding.

3.4.7 Upholstery fabric attachment. The upholstery fabric shall be in not more than three pieces with seams allowed near the perimeter edge, between the seat and back and, when three pieces are used, approximately half way up the back. The seat and back of the type IV chair shall each have one piece of upholstery fabric. The buttons used in tufting shall be covered with the same fabric used in upholstering the chair. The buttons shall be tied through the upholstery fabric and padding and through apertures in the inner shell and held securely so that they cannot pull out. There shall be no less than four buttons in the seat and no less than four buttons in the back except type IV chairs which shall have no less than two buttons in the back. The buttons shall be symmetrically spaced, compatible with the contour of the seat and back. The upholstery fabric shall be turned under the perimeter of the inner shell and securely fastened to prevent unintentional removal. There shall be sufficient internal structure to hold the fabric taut to prevent the upholstery cover from slipping. The chair shall be neatly upholstered, having a finished appearance without wrinkles, puckers, or sags. The inner shell shall be capable of being removed from the outer shell, reupholstered and then reattached to the outer shell.

3.5 Pretreatment and finishing.

3.5.1 Paint pretreatment. All unplated ferrous metal surfaces, (except the spindle), shall be treated for painting in accordance with the manufacturer's commercial practice.

3.5.2 Paint finishing. All pretreated ferrous metal surfaces shall be coated with enamel. The color shall be black. The finish shall level out to a smooth, uniform surface. There shall be no areas of thin film, metal showing through the finish or separation of color.

3.5.3 Plating pretreatment. The components to be plated shall be cleaned and otherwise pretreated in accordance with the standard commercial practice for the type of plating specified.

3.5.4 Chromium plating. The exposed surface of the chromium shall visually match Government sample number FSS-M-01001 (see 6.3). The chromium plating shall not peel, chip, or crack and shall be free of uneven color, pits, blisters, stains, powder film, and brush or grind marks. The chromium plated steel shall pass the test in 4.6.9.

3.6 Identification label. The identification label shall be rectangular, minimum 2-1/2 inches by 1-1/2 inches, legibly marked in permanent dark ink with the contractor's name or trademark, contract or order number, item stock number and month and year of manufacture. It shall be affixed to the underside of the chair with a permanent type adhesive, so that after being applied for 4 hours it can not be removed by hand without defacement.

3.7 Adjustment instructions. Each chair shall have a printed tag which clearly shows each adjustment point of the chair and how adjustment is affected. The tag shall be attached to part of the chair's adjustment mechanism. The printing shall be legible and permanent.

3.8 Workmanship. A high degree of craftsmanship shall be exercised while performing the various operations of manufacture to produce a finished commodity that is satisfactory in function, serviceability, and appearance. There shall be no burrs, sharp edges, or other defects that could be hazardous to persons or clothing. The finished chair shall function properly in all respects and shall not squeak or make annoying noises in use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

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4.2 Preproduction tests. Tests shall be conducted under the supervision of the Government Inspector at the manufacturer's plant. In the event the manufacturer does not have the facilities to perform the required tests, he shall engage the services of a commercial testing laboratory acceptable to the Government. Copies of the laboratory report shall be submitted to the Government Inspector upon receipt from the laboratory. All examination and testing shall be in accordance with section 4 of this specification. Notwithstanding the above, the government reserves the right to perform preproduction testing at its own or a contracted facility.

4.3 Sampling for inspection and acceptance. Sampling for lot inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.4 Inspection (end item).

4.4.1 Visual examination. Examination of the end item(s) shall be in accordance with the classification of defects specified in table III and acceptable quality levels (AQLs) specified in 4.4.3. The lot size shall be expressed in the number of completed units for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one complete chair.

4.4.2 Dimensional examination. Inspection will be made of chairs for compliance with dimensions specified. Any dimension not within tolerance specified shall be classified as a major defect. The points at which the dimensions are to be taken are as set forth in figure 5.

4.4.2.1 Tilt range measurement. The range of travel of the tilting mechanism is determined by taking angle measurements of the tilting mechanism with the tension completely released.

TABLE III. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Casters/Glides (3.2.5.2, 3.2.5.3)	Not type or style caster specified.	X	
	Not type of glide specified.	X	
	Missing	X	
	Tread of caster not proper hardness.		X
Chair controls (3.2.5.4)	Tilting range not as specified.		X
	Seat adjustment range less than 2-1/4 inches.	X	
	Seat height changes when chair revolves.	X	
	Tilting unit not provided with positive safety stop.	X	
	Adjustment instructions missing.		X
Welding and brazing (3.4.1, 3.4.2)	Welding and brazing not as specified.	X	
	Burnt through, cracked, fractured or not fused.	X	
Base (3.4.3, 3.4.3.1)	Caster/glide sockets not as specified.	X	
	Not design specified.		X
	Not material specified.	X	
	Not finish specified.		X
	All legs not uniform in slope and position on hub.	X	
Shells (3.4.4)	Material not as specified.	X	
Arms (3.4.5)	Attaching brackets, when used, are visible.		X
	Color not incorporated at time of molding.		X
Padding (3.2.2, 3.4.6)	Not material specified.	X	
	Not thickness specified.	X	
	Not secured to inner shell.		X
Upholstery fabric (3.2.3, 3.4.7)	Not material specified.	X	
	Faded or shade variations affecting appearance.		X
	Tailored improperly affecting appearance.		X
	Not securely attached.	X	
Chromium plating (3.5.4)	Plating not as specified.	X	
	Plating finish not as specified.		X
Identification label (3.6)	Missing		X
	Wrong location.		X
Workmanship and construction (3.8)	Production unit varies from accepted preproduction sample.	X	
	Malformed, damaged or warped parts, burrs, and sharp edges affecting appearance, function, or safety.	X	
	Components not of specified material.	X	

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4.4.3 Inspection levels and AQL's. The acceptable quality levels (AQL's) expressed in percent defective and inspection levels shall be as follows:

	Inspection levels	AQL's
For examination in table III Major	II	4.0
For examination in table III Total	II	10.0
For examination in 4.4.2	S-2	4.0

4.4.4 Certification for recovered material. A certificate shall be submitted, stating that the end item(s) supplied contain the percentage of recovered material required herein (see 3.2).

4.5 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container identical to each unit being shipped fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.6 Tests. The following tests shall be performed as described herein on each type and style of chair being produced under the contract. Tests shall be performed on the preproduction sample chairs and component parts submitted for inspection and thereafter as deemed necessary by the Government. One preproduction sample chair shall be used to perform the tests described in 4.6.5 and 4.6.11. The other preproduction sample chair and component parts shall be used to perform all remaining tests. Acceptance of chairs shall be determined by checking for the following defects during and after the required testing: Structural breakage or loss of serviceability; cracks or fractures in the base, arms, or controls; loosening of fastening devices; inoperative chair controls; or failure that in any way could cause personal injury. At the completion of the required tests, the chairs shall be completely disassembled and examined for cracks or fracture in the inner and outer shells and all of the above listed defects. Failure to meet the requirements specified herein including occurrence of the above listed defects shall be cause for rejection of the type and style chair being examined and tested.

4.6.1 Drop test, for all types of chairs (figures 6 and 6A). Chairs with height adjustment shall have the seat height set at 1-1/2 inches above the minimum adjustment. Casters or glides shall be removed. Two adjacent arms of the base shall be secured to prevent movement during the test. Make reference measurements at both front corners of the seat underside for determining permanent distortion. A bag of sand or shot 12 to 16 inches in diameter, weighing 275 pounds for type I, II, and III chairs and 200 pounds for type IV chairs shall be dropped once onto the center of the seat from a height of 6 inches. Permanent distortion shall not exceed 1/4 inch.

4.6.2 Back pull testing for type I, II, and III chairs (figure 7). Casters shall be removed and seat height set at midpoint. The chair shall be secured to the platform in an upright position by means of clamps on the base arms and shall be restrained from movement. Chairs with tilting units shall have the tilt tension set at the middle of the adjustment range. A spring dynamometer or equally effective device shall be attached to a yoke or clamp and fastened to the chair back 16 inches above the seat or as close to the top of the chair back as practical when the dimension from the seat to the top of the chair back is less than 16 inches. Apply a 5 pound load then make initial reference measurements from the yoke attachment point for determining permanent distortion. A back pull of 250 pounds shall be applied normal to the slope of the back and maintained for not less than 3 minutes. Permanent distortion shall not exceed 1 inch.

4.6.2.1 Back pull test for type IV chairs (figure 8). With casters removed, restrain the base of the chair from movement. The seat height shall be set 1-1/2 inches above the minimum seat height adjustment and the back height and horizontal adjustment shall be set at the medium position. A spring dynamometer or an equally effective device shall be attached to a yoke fastened to the center of the back rest, and a back pull shall be applied normal to the slope of the back at all times and up to the maximum limit of travel. A preload back pull of 50 pounds shall be applied for a period of 1 minute then all but 10 pounds of pull shall be released. Make initial reference measurements from the yoke attachment point for determing permanent distortion. The back shall then have a test load pull of 150 pounds applied for a period of 3 minutes, then all but 10 pounds of pull released and the permanent distortion measured. Permanent distortion shall not exceed 1/4 inch. Measurements for determining distortion shall be taken on a horizontal line from point of attachment before and after the application of the test load with the chair back in its normal upright position.

4.6.3 Base test, static load, for all types of chairs (figure 9). The casters or glides shall be removed and the base shall be placed in an upright position, supported with steel blocks under the base legs as shown in figure 9. A static load of 2,500 pounds, concentrated over an area of 1-3/4 inches in diameter at the center of the base, shall be applied at the rate of 0.10 to 0.125 inch per minute. The base shall withstand the test without any permanent distortions exceeding 1/16 inch measured from a point on the metal center hub (not a plastic liner), structural failure, or interior or exterior breakage.

4.6.4 Arm deflection test for style A chairs (figure 10). Take a reference measurement between the arms 1 inch back from the vertical upright of the arm to determine arm deflection. Apply equal and opposite horizontal pulls of 40 pounds outward on each arm simultaneously at the most forward point of the arm rest. Deflection between the arms under the 40 pound loads shall not exceed 1-1/2 inches. Increase the horizontal loads to 175 pounds on each arm simultaneously. Permanent distortion between the arms after release of the 175 pound loads shall not exceed 1/2 inch.

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4.6.5 Durability test for casters and base for type I, II, and IV chairs. A chair with casters mounted shall be placed on the obstacle layout as indicated in figure 11. A 300 pound load shall be applied (as indicated in figure 11) with the spindle fully extended. The base shall be attached to a mechanical device which will exert a horizontal push and pull of from 30 inches to 34 inches as illustrated in the figure. The machine will operate continuously at a rate of 8 to 10 cycles per minute with a maximum speed of 50 feet per minute. One cycle consists of a forward and backward stroke of the mechanical device. The base will be cycled for 36,000 cycles. When there is caster failure within acceptable limits specified below, the casters shall be replaced and the test continued. Any of the following failures shall be cause for rejection:

(a) Structural failure, raceway failure, seperation of tread or excessive tread wear on more than one caster within the first 20,000 cycles of testing.

(b) Any failure described in (a) above in more than a total of two casters before the completion of 36,000 cycles.

(c) Any structural failure or internal or external breakage of the base at any point in the test.

4.6.6 Off-center test for chair support assembly for all types of chairs (figure 12). Each chair control mechanism with spindle attached shall be secured to a testing platform. The spindle shall be clamped 1-1/2 inches from spindle bottom. A loading frame shall be attached to the spider. A vertical force of 680 pounds concentrated over an area of 1 inch shall be applied at a rate of 0.10 to 0.125 inch per minute onto the center of the loading frame 6 inches forward of the centerline of the spindle. The chair support assembly shall withstand the loading with no mechanical or material failure of any part of the assembly and a maximum allowable permanent distortion of 1/8 inch as measured from the point of application of the load.

4.6.7 Spindle attachment test. The spider, chassis, and spindle shall be detached from the chair and tested as follows:

(a) Apply a torque of 100 foot pounds to turn or displace the spindle by use of a tool, such as a wrench.

(b) Apply a force of 1,500 pounds to the spindle in the direction of its removal from the control at the rate of 0.10 to 0.125 inch per minute. The spindle shall remain unmoved after the completion of the tests.

4.6.8 Test for upholstery fabric. The upholstery fabric shall be tested in accordance with the requirements of 3.2.3 and any failure to comply with the requirements when tested in accordance with the test methods cited shall be reason for rejection.

4.6.9 Chromium plating test. Test samples shall be plated as specified in 3.5 and shall be tested for corrosion in accordance with the 5 percent salt spray test at 95 degrees F in ASTM Test Method B 117. The test period shall be not less than 48 hours. After the specified exposure is complete, the test specimen shall be visually examined to assure that the plating shows no evidence of peel, chip, uneven color, powder film, pits, or stains. Widely scattered spots of corrosion are permitted, however, none of the spots of corrosion shall exceed 1/16 inch along the greatest dimension. Bleeding and rusting above welds and seams in normally unexposed areas should be disregarded.

4.6.10 Adhesion test for polyester finish. When the outer shell is colored with a baked on polyester finish, the finish shall be subjected to the following adhesion test. A sample of the outer shell shall be prepared as specified in 3.4.4.1. The samples shall be scored through the finish to the shell with a razor blade in such a manner as to produce a grid of a minimum of 100 1/16 inch squares. The dried film shall not be removed from the sample when a 1 inch wide piece of cellophane tape (Scotch Brand No. 600 or equal) is applied firmly to the grid surface (full length of panel) and then is quickly pulled from the surface.

4.6.11 Swivel cycling test for all rotary base chairs (figure 13). Set seat height adjustment to midpoint on adjustable chairs. The base of the entire chair shall be clamped to a rotating platform. A 225 pound load shall be secured on the seat 2 inches forward of the center line of the spindle as shown in figure 13. The base shall be rotated a complete 360 degree cycle (except type III chairs with return swivel controls shall be rotated in a reciprocating movement from stop to stop) for 120,000 cycles at a frequency of 10 to 16 cycles per minute. Structural failure at any point in the test, failure of the seat to swivel without dragging or binding, or failure to comply with the test in 4.6.14 shall be cause for rejection.

4.6.12 Seating impact test for all types of chairs. The chair shall be secured to a test platform. Chairs with adjustable features shall have all adjustments set at mid-point of adjustment. Chairs with casters shall have casters perpendicular to base legs. A cycling machine shall drop a 125 pound bag of sand or shot 12 to 16 inches in diameter from a height of 2 inches above the seat onto the center of the seat. The drop shall be cycled for 100,000 cycles at a speed of 20 + 5 cycles per minute. Any evidence of structural failure in any of the materials, cracked or broken welds, broken casters, casters forced out of square with base more than 1/4 inch, damage to height adjustment mechanism or failure to comply with the test in 4.6.14 shall be cause for rejection.

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4.6.13 Tilting mechanism back stop and durability test. The test shall be performed on the type I, II, and IV chairs. The test may be performed on the entire chair or alternatively, on the chair without the upholstery. The chair shall be secured to a test platform with casters removed. The chair controls shall be set at mid-point of adjustment. A cycling device shall be attached to the chair back (front push or back pull) so that the load is applied 16 inches above the seat; or to the top of the back frame if the frame is lower than 16 inches from the seat; or to the pivot point of a pivoting back. The cycling device shall apply a 100 pound force to the chair back of the type I and II chairs and a 75 pound force to the back of the type IV chair. The force shall be normal to the plane of the back at the back stop position. A weight of 100 pounds shall be secured to the center of the chair seat. The device shall cycle the the chair for 120,000 cycles at a constant frequency between 10 to 40 (10% variance) cycles per minute. Structural failure in any of the materials, cracked or broken welds, damage to tilt control mechanism, or failure to comply with the test in 4.6.14 shall be cause for rejection.

4.6.14 Measurement of excessive play (wobble) between chair seat and base (applicable to 4.6.11 through 4.6.13). Before the chair is taken apart, the chair shall be checked for excessive wobble between the base and control mechanism. The casters or glides shall be removed and the chair shall be placed on a level surface and visually examined to assure all base legs contact the level surface so there is no wobble between the chair base and the level surface. If any leg is off the level surface more than 1/4 inch the chair shall be rejected. The chair shall then be secured to a level surface plate with the seat height and control adjustments set at maximum height. A weight of 15 pounds shall be suspended from the center right side edge of the chair seat. Measurements shall be taken from the level surface plate to the pre-established reference points on the front-to-rear center of both sides of the seat. The weight shall then be suspended from the opposite side and the measurements retaken. The difference between the two left side measurements and the two right side measurements shall be calculated. The chair seat shall then be rotated 90 degrees and the entire set of measurements retaken and the difference between the two left side measurements and the two right side measurements calculated. The average of all four differences shall be calculated. This average shall not exceed 3/16 inch.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Levels A and B. Each chair shall be secured in its lowest position, if adjustable. When casters are removed they shall be packaged in a cloth or polyethylene bag and secured with reinforced pressure-sensitive adhesive tape to the rotary mechanism or base of the chair to prevent loss or damage during multiple shipment, handling, and storage. All finished surfaces of the chair that come in contact with the inside of the shipping container shall be covered with cushioning material which shall be secured in place with tape or twine. The cushioning material shall extend not less than 2 inches above, below and to each side of the point of contact. The cushioning material shall be in the form of pads which shall be made of any of the following materials: macerated paper, wood wool batting, wood fiber felt or crepe cellulose wadding. The cushioned chair shall then be wrapped in a polyethylene bag or shroud a minimum of 1 mil thick extending a minimum of 6 inches below the bottom of the seat to protect the upholstered surfaces. In lieu of the polyethylene bag and protective pads, the chair shall be covered with a cushioned bag extending a minimum of 6 inches below the bottom of the seat. The bag shall be constructed from material conforming to Item 42[illegible], Note 4, of the Uniform Freight Classification.

5.1.2 Commercial. Each chair shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the chair against damage during multiple shipments, handling, and storage, while complying with National Motor Freight Classification and Uniform Freight Classification, as a minimum.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each chair packaged as specified in 5.1.1 shall be packed as specified in 5.2.2 except that the boxes shall conform to PPP-B-636, style FOL, grade W3C. Closure, water-proof sealing, and reinforcing of the boxes shall be in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Each chair, packaged as specified in 5.1.1 shall be packaged in a close-fitting box conforming to PPP-B-636, style RSC or FOL, class domestic, variety DW, grade 275 minimum. The chair shall rest on a pad, which is of the full inside dimensions of the box and of the same material as the box. Additional blocking and bracing shall be utilized throughout the container as required to prevent movement or damage during multiple shipments, handling, and storage. The boxes shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Commercial. Each chair, packaged as specified in 5.1.2, shall be packed in a fiberboard box to insure delivery at destination and provide protection during multiple shipments, handling, and storage. The pack shall be acceptable to the common carrier in an approved "F" pack for chairs of this type, under National Motor Freight Classification and Uniform Freight Classification, as a minimum.

5.3 Marking. Marking shall be as specified in the contract or order.

5.3.1 Special marking. In addition to the markings required by 5.3, levels A and B shipping containers shall be marked as follows, in characters proportionate to the box size with a minimum height of 3/4 inch.

One side and one end of each container:

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The top panel of each shipping container:

TOP

THIS SIDE UP

FRAGILE - HANDLE WITH CARE

The upper corner of each side and end panel shall be marked with the word "UP" in characters a minimum of 2 inches high, and an arrow pointing to the top of the container.

6. NOTES

6.1 Intended use. The specification covers chairs suitable for use at desks and in conference rooms. They are modern in design and provide quality and comfort.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

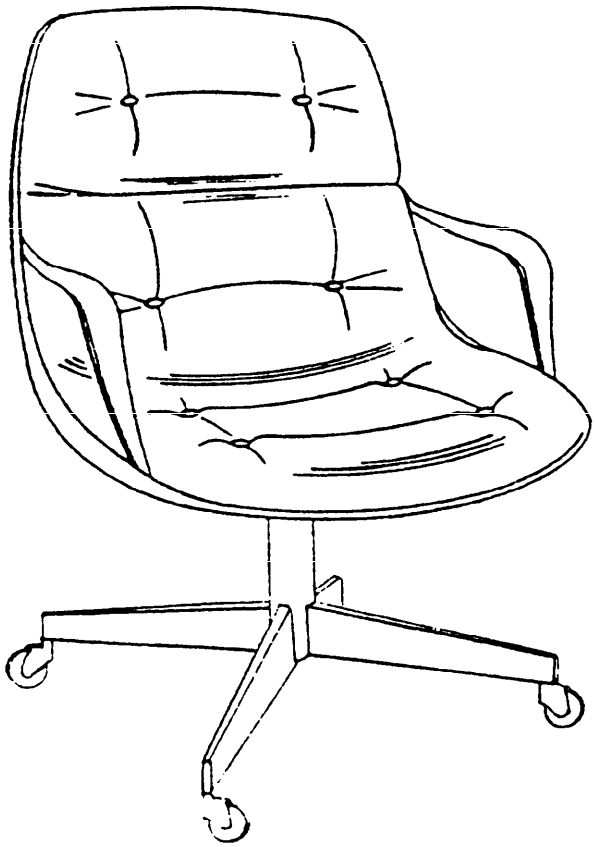
(a) Title, number, and date of this specification.
(b) Type and style of chair required (see 1.2.1).
(c) Color of upholstery fabric desired (see 3.2.3.1).
(d) Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Standard samples. Samples of the standard colors of upholstery fabric and satin chrome finish are obtainable from the Furniture Branch, Office of Engineering and Technical Management, Federal Supply Service, General Services Administration, Washington, DC 204066.

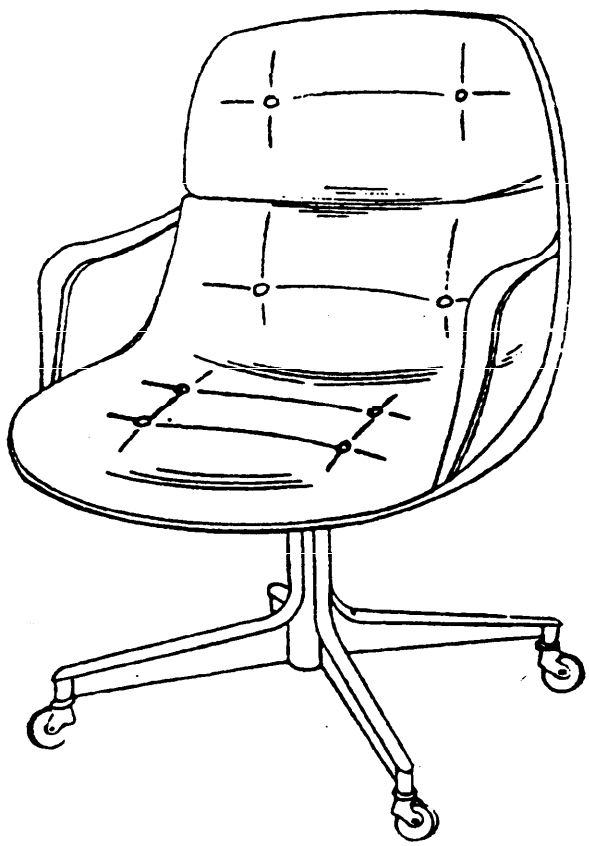
6.4 Warranty. When specified in the contract, a contractor warranty shall be provided for the inner and outer plastic shells which includes, as a minimum, the following elements:

(a) Period of 2 years to begin 60 days after date of manufacture on the identification label.
(b) Free from cracks, fractures, discoloration, breakage, distortion, peeling, molding defects, and defects which could cause personal injury.
(c) Contractor will replace or repair at no cost to the Government.

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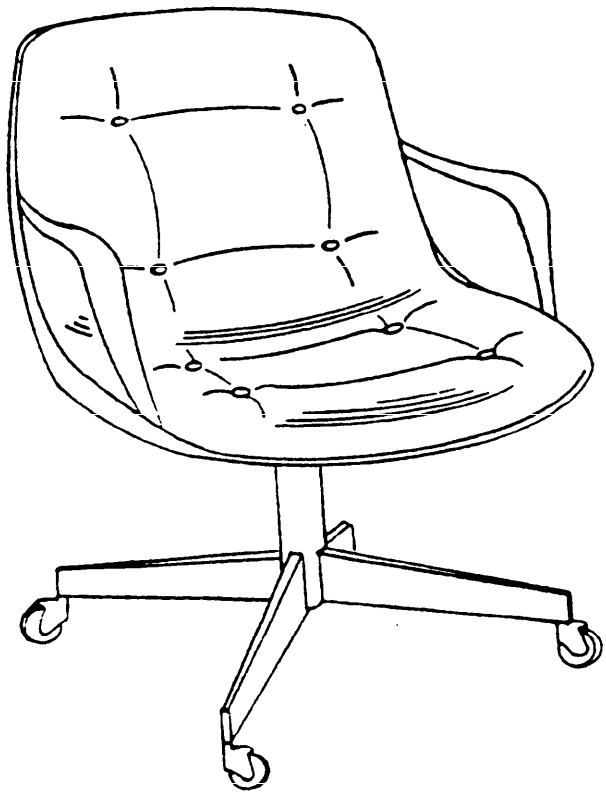
Type I, Style A



Type I, Style A

FIGURE 1

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Type II, Style A



Type II, Style B

Figure 2

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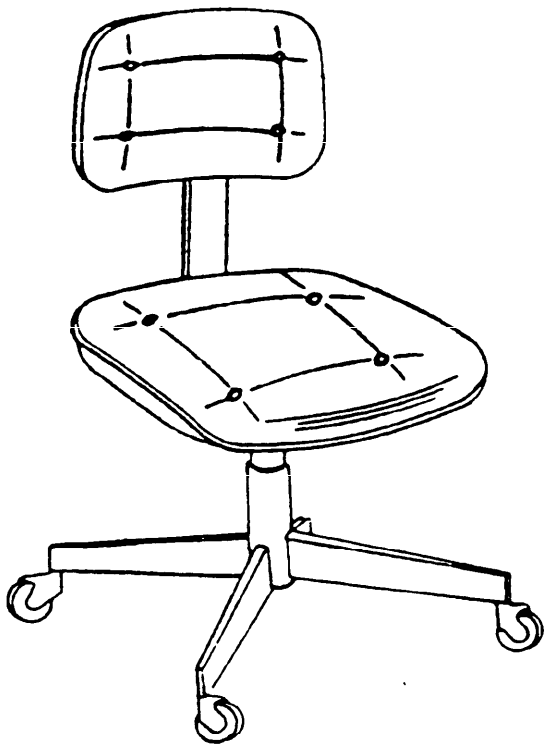
Type III, Style A



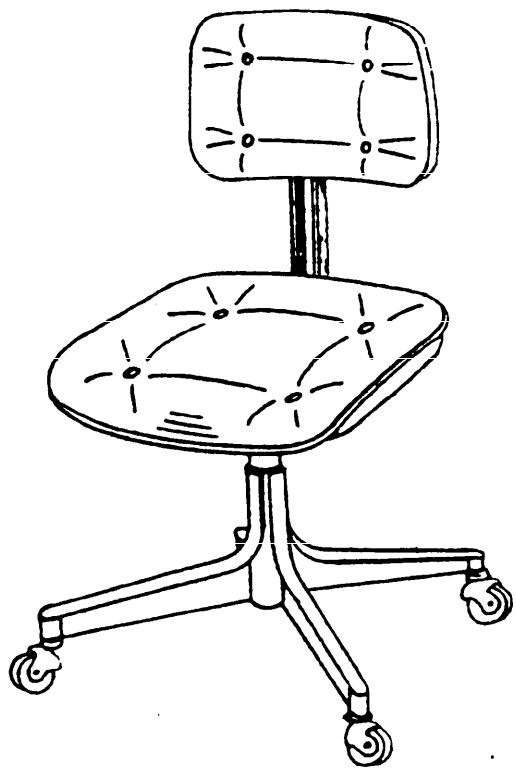
Type III, Style B

Figure 3

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Type IV, Style B



Type IV, Style B

Figure 4

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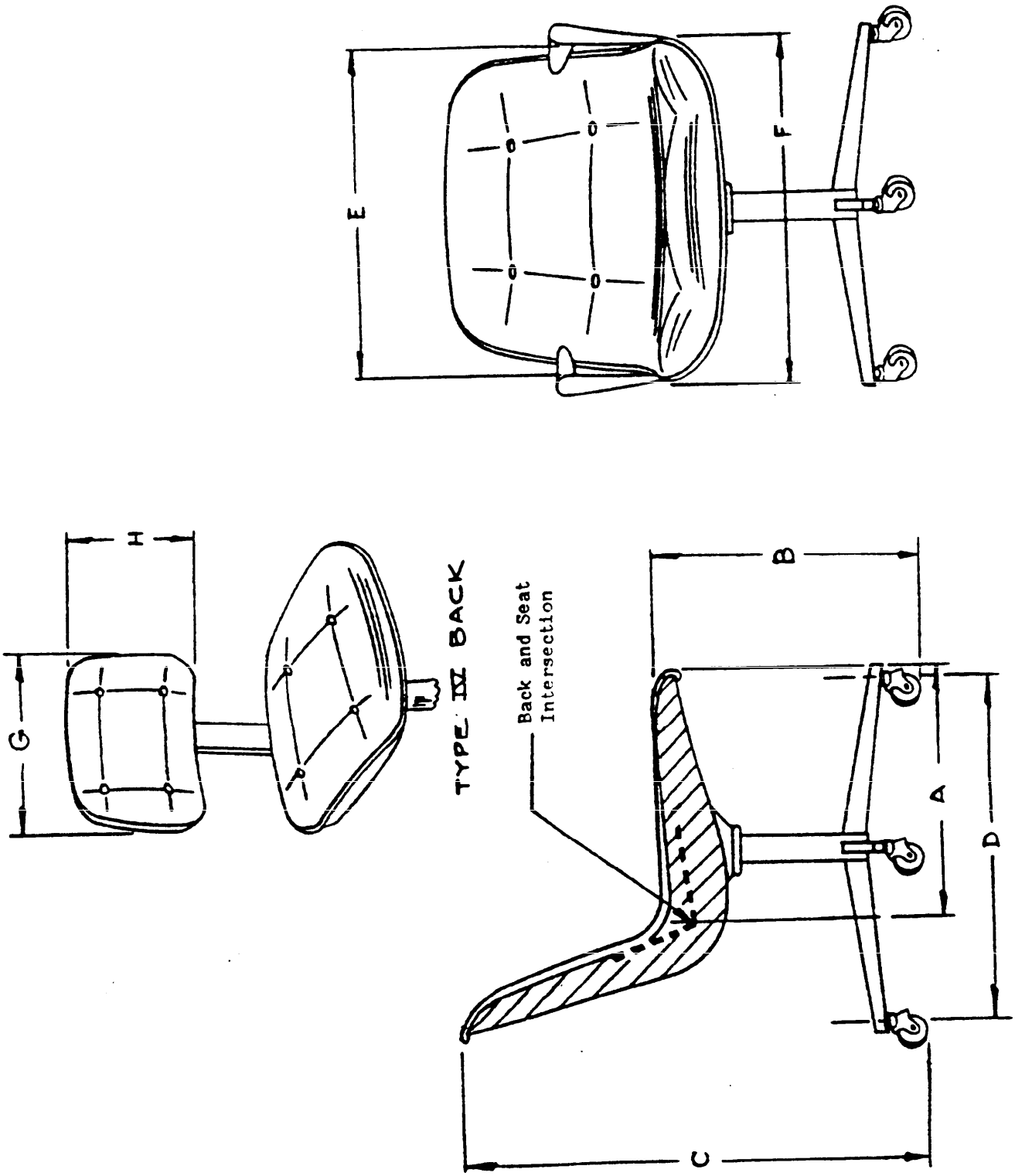


Figure 5

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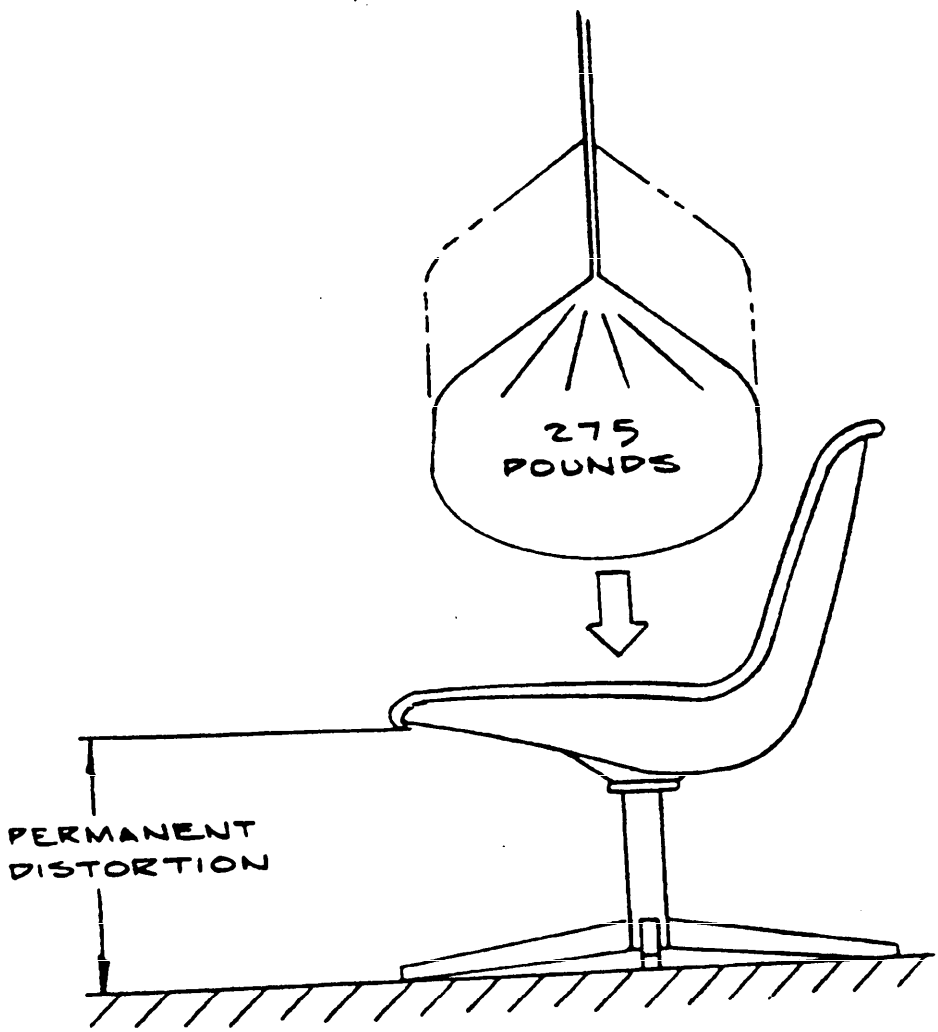


Figure 6

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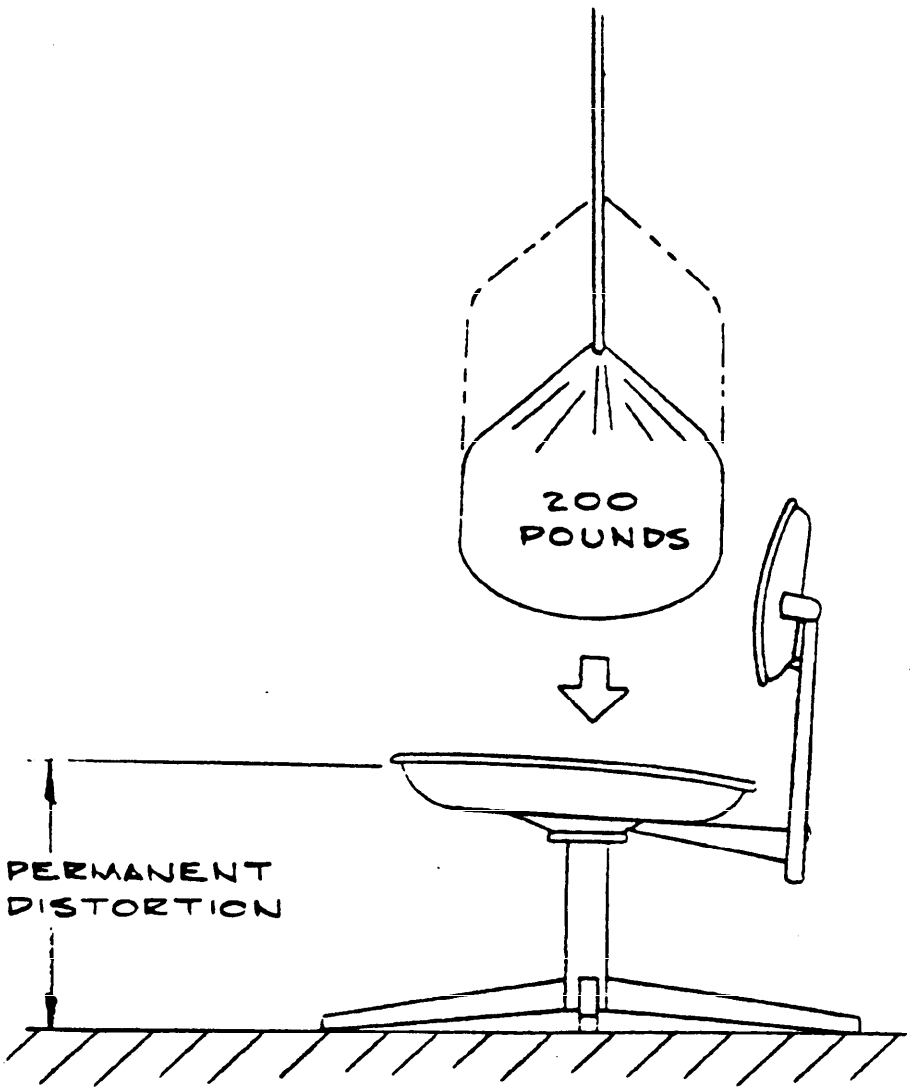


Figure 6A

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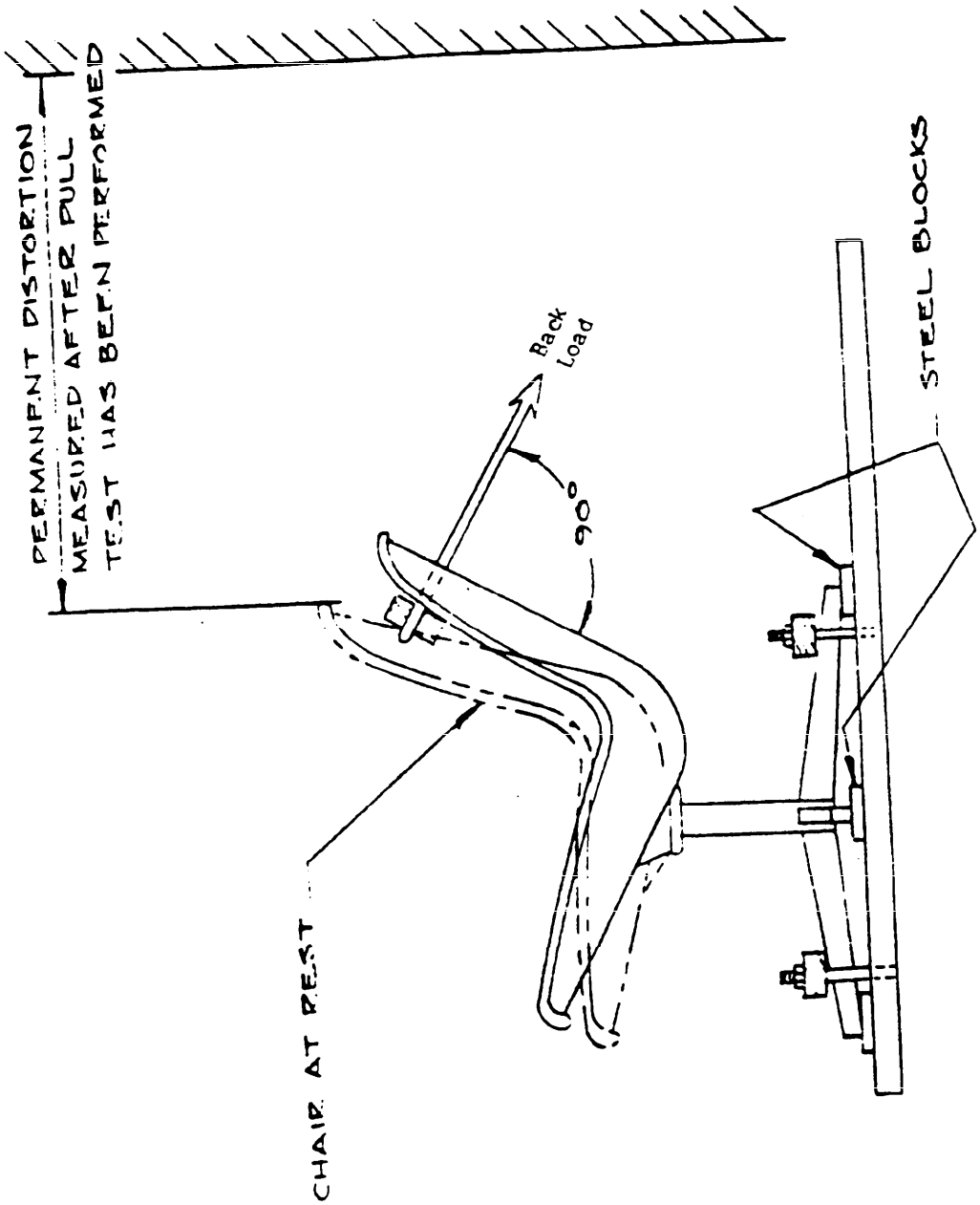


Figure 7

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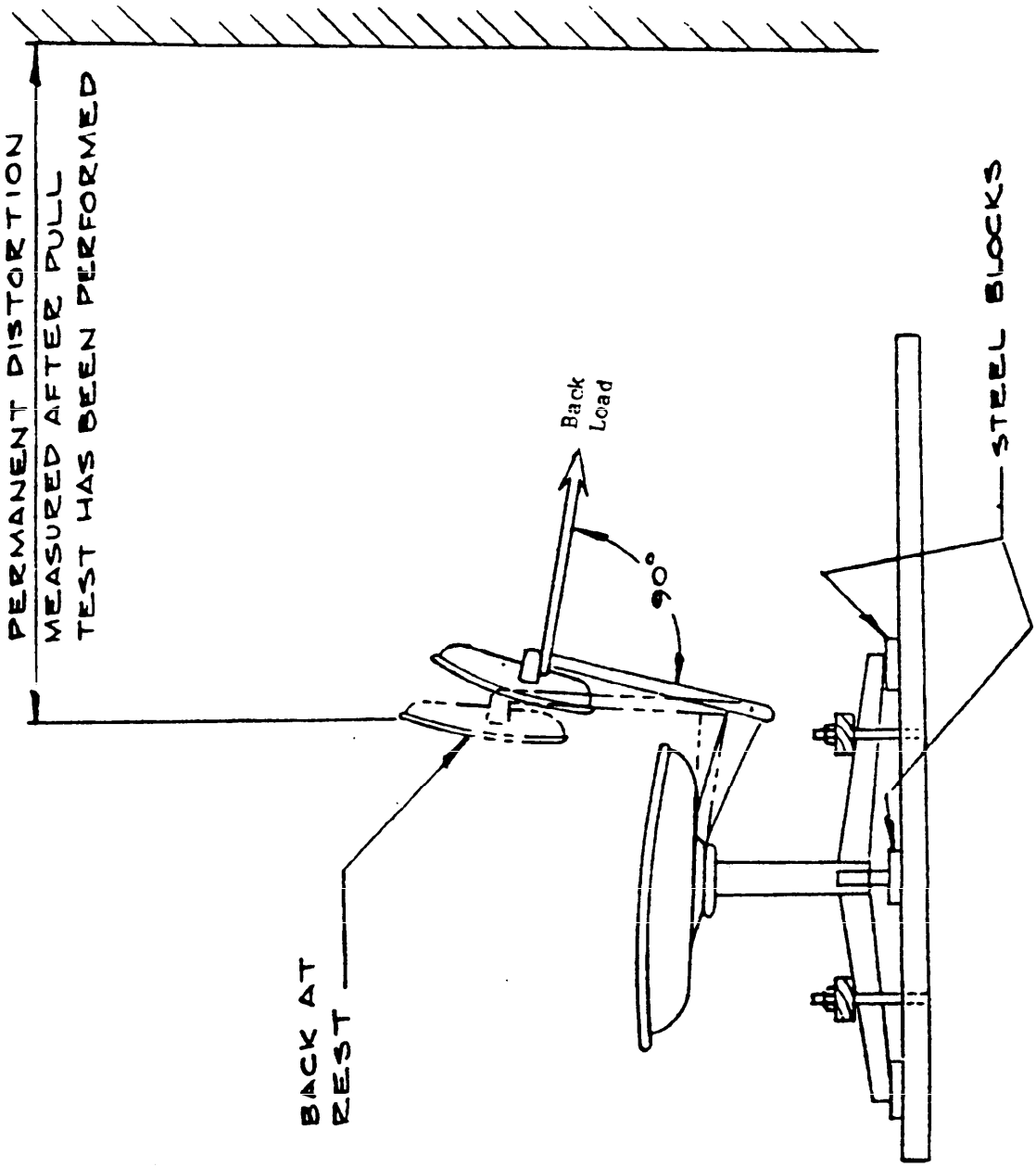


Figure 8

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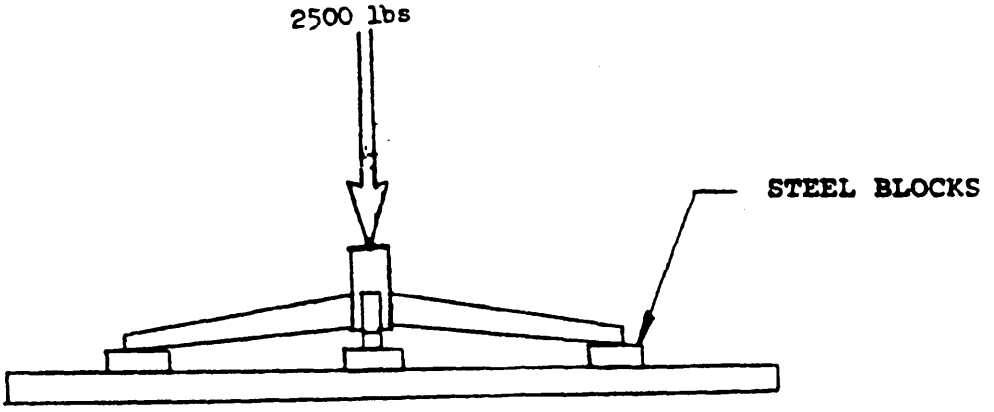


Figure 9

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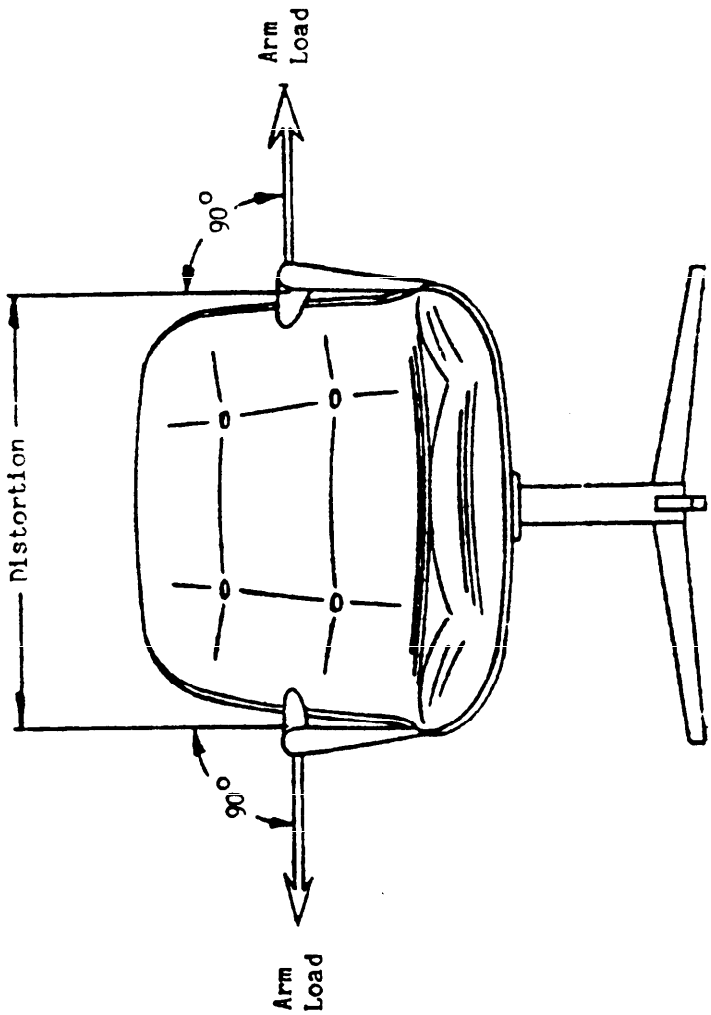


Figure 10

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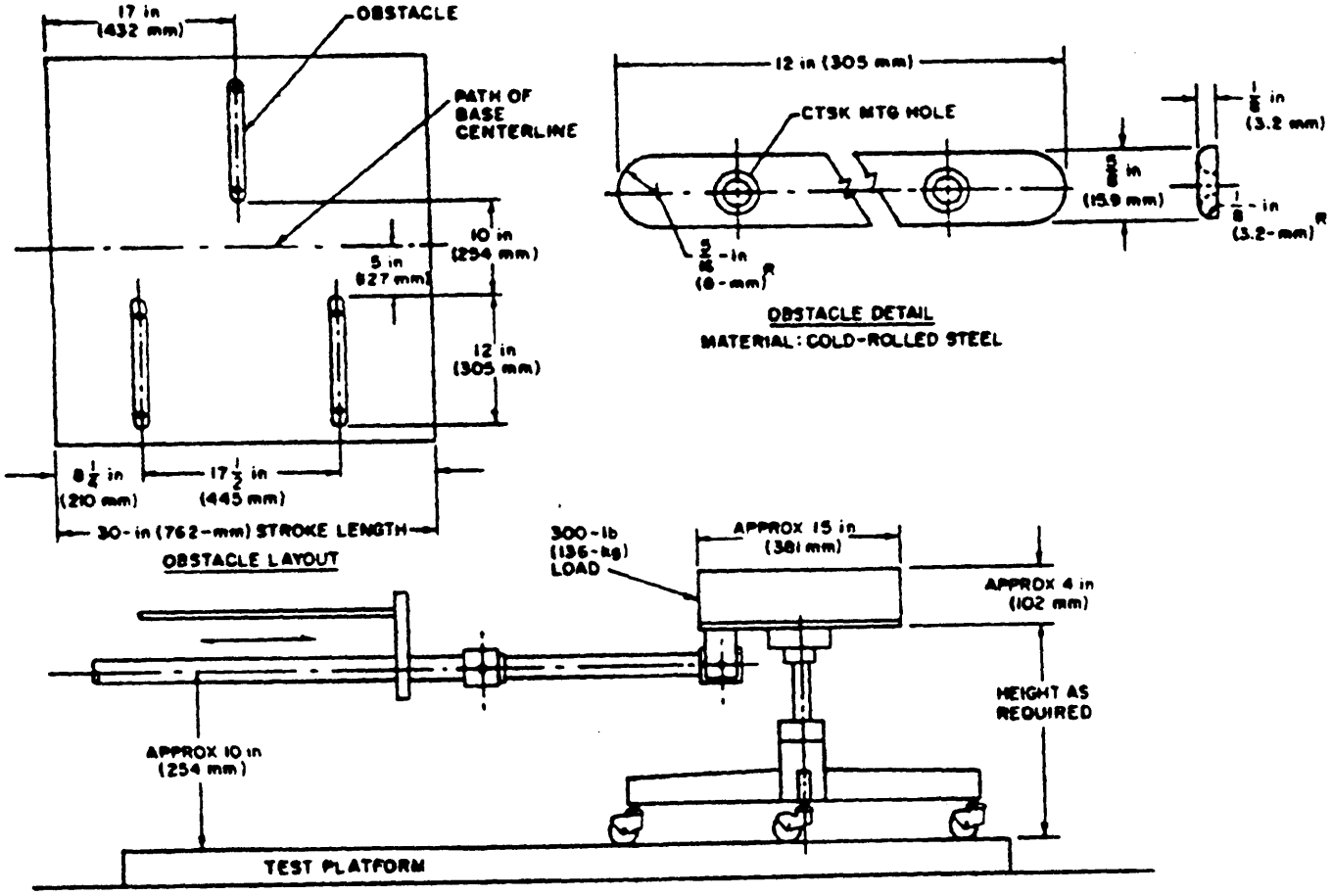


Figure 11

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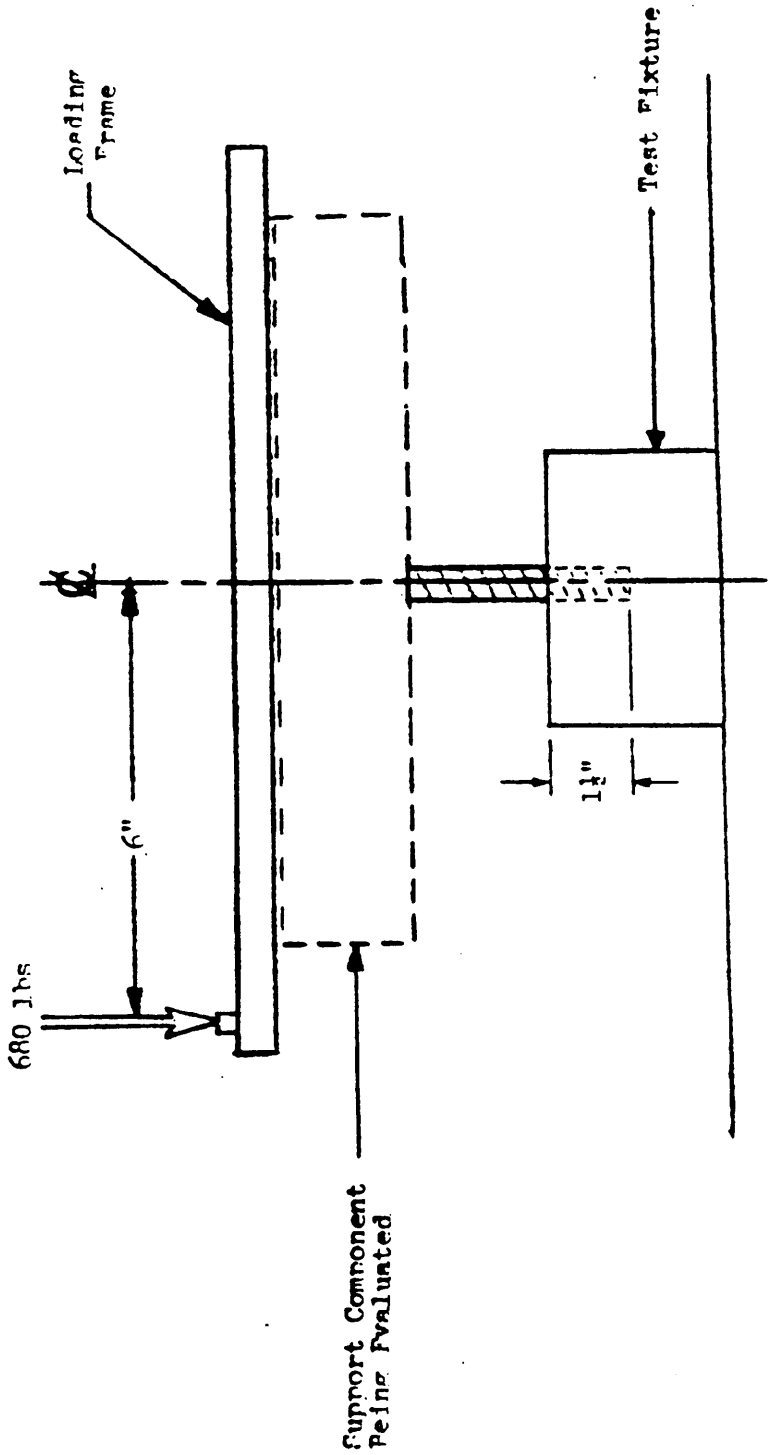


Figure 12

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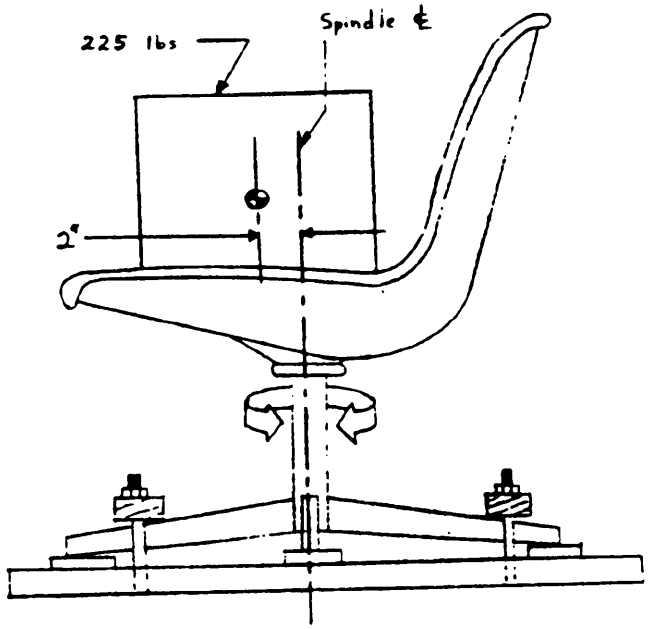


Figure 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:41Z
Check the source to verify that this is the current version before use.